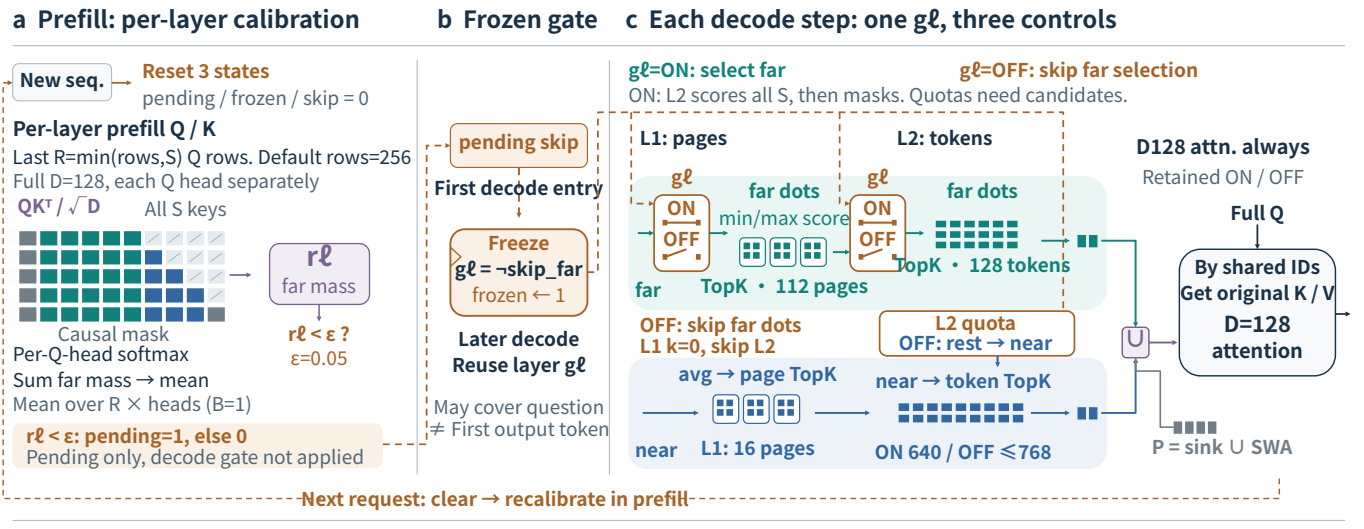

a Prefill: per-layer calibration
b Frozen gate
c Each decode step: one gℓ, three controls
New seq.
Reset 3 states
pending / frozen / skip = 0
Per-layer prefill Q / K
Last R=min(rows,S) Q rows. Default rows=256
Full D=128, each Q head separately
QKᵀ / √D
All S keys
Causal mask
rℓ
far mass
rℓ < ε ?
ε=0.05
Per-Q-head softmax
Sum far mass → mean
Mean over R × heads (B=1)
rℓ < ε: pending=1, else 0
Pending only, decode gate not applied
pending skip
First decode entry
Freeze
gℓ = ¬skip_far
frozen ← 1
Later decode
Reuse layer gℓ
May cover question
≠ First output token
gℓ=ON: select far
ON: L2 scores all S, then masks. Quotas need candidates.
gℓ=OFF: skip far selection
L1: pages
L2: tokens
gℓ
ON
OFF
far dots
min/max score
TopK • 112 pages
far
TopK • 128 tokens
OFF: skip far dots
L1 k=0, skip L2
L2 quota
OFF: rest → near
avg → page TopK
near → token TopK
near
L1: 16 pages
ON 640 / OFF ⩽768
U
D128 attn. always
Retained ON / OFF
Full Q
By shared IDs
Get original K / V
D=128
attention
P = sink ∪ SWA
Next request: clear → recalibrate in prefill